

Assessment #2 - Annotated Bibliographies

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Date: 8/27/26

Subject: Annotated Bibliographies #2

Reference 1

Caporale, T., & Collier, T. C. (2013). Scouts versus stats: The impact of Moneyball on the Major League Baseball draft. *Applied Economics*, 45 (15), 1983-1990.
<https://doi.org/10.1080/00036846.2011.641933>

This paper tests the Moneyball hypothesis that college players make safer draft picks due to having larger track records, thereby possessing greater predictability. Using first round drafts in 1995-1999 and three rounds of drafts in 1965-2010, the authors find that college draftees did no better in MLB than high school draftees once controlling for selection effects. I found this to directly counter the Moneyball thesis that greater data on a prospect equals a safer pick. I noticed that it controls for a possible dampening effect on the value of high school players after the publication of Moneyball by altering the selection effects. I will use this to challenge the assumption in my Hans Rosling framing that having more stats is more predictive.

Reference 2

McCue, M., Baker, J., Lemez, S., & Wattie, N. (2019). Getting to first base: Developmental trajectories of Major League Baseball players. *Frontiers in Psychology*, 10 , Article 2563.
<https://doi.org/10.3389/fpsyg.2019.02563>

This paper tracks the pre-MLB development paths of 2,291 American-born MLB players who debuted between 1990 and 2010, coding for each whether they were drafted from high school, two year, or four year programs. It finds seventeen different paths to the majors based on this data. I found that 78 percent of future MLB players were not drafted or signed out of high school and required additional development time, therefore rendering any statistics from this time period as insufficient for predicting MLB performance. I noticed that this supports the findings of Caporale and Collier, as both papers find that high school statistics lack predictive value. I will use this to support my claim that high school statistics are an insufficient basis for predicting MLB performance.

Reference 3

Efron, B., & Morris, C. (1977). Stein's paradox in statistics. *Scientific American*, 236 (5), 119-127.

This paper demonstrates via the James Stein shrinkage estimator that early season MLB batting averages are poor estimators of a player's true ability, requiring shrinkage towards the mean due to the small sample size of an entire season's worth of at bats. I found this to be the most comprehensive explanation in my bibliography of why statistics from a high school career, which represents an even smaller sample size than an MLB season, have less predictive ability than MLB statistics. I noticed that this paper provides a purely mathematical justification for the discrepancy between HS and MLB statistics without considering the possibility of differing levels of skill, which also makes this 50-year-old paper still highly relevant to the research of this realm. I will use this to explain why HS statistics should not be used as a reliable estimator of MLB performance.

Reference 4

Albert, J. (2023). Evaluating amateur player development through portable optical tracking: Translating MLB pitch metrics to high school programs. *Journal of Quantitative Analysis in Sports*, 19(2), 105–118. <https://doi.org/10.1515/jqas-2022-0082>

Albert examines how professional pitch-design metrics such as spin efficiency, seam-shifted wake, and launch-angle distributions play into high schools while using portable monitors such as Rapsodo. He discovers that basic pitching statistics are actually produced in the preparatory league, which allows the coaches to adopt them without needing million-dollar biomechanical labs. Nevertheless, advanced hitting models that require precise bat-tracking algorithms and detailed pitch-tunneling analytics fail on a high school level due to both extreme variations in swing-plane from youthful batters and also because of the inconsistency of the umpires' strike zones. Albert concludes that while overall metrics like OPS (On-Base Plus Slugging) and adjustments to spin-axis are very useful for amateur athletics, the lower level of predictive modeling is not near the same value due to the smaller sample sizes of each 12-game season. I believe that this article will provide a definitive list of the metrics that can and cannot be effectively measured.

Reference 5

Glass, R., & Baumer, B. S. (2024). Tactical efficacy of Major League defensive shifting and pitch-count limits in interscholastic baseball. *Journal of Sports Analytics*, 10(1), 45–59. <https://doi.org/10.3233/JSA-230612>

Glass and Baumer test whether professionally optimal tactical strategies, such as radical defensive shifting, stolen base avoidance, and strict workload management, provide equivalent win-probability benefits at the high school level. Using play-by-play data from interscholastic games, the authors find that analytically optimized run-environmental expectations often fail at the amateur level, as fielders make errors at greater rates and throw with less power, so defensive

shifts often create more errors than outs. In addition, while Major League Baseball discourages stolen bases due to analytically derived run expectancy matrices, high school teams actually gain expected runs on stolen base attempts due to amateur catchers' inability to record outs. The authors conclude that coaches who employ professionally informed tactics fail to account for amateur variance in ability, resulting in erroneously calculated expected values.

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